



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

ANGULAR MEASUREMENTS

LSVMA302

MEASURE ANGLES

Competence

RQF Level: 3

Learning Hours



60

Credits: 10

Sector: Construction and Building Services

Trade: Land Surveying

Module Type: Specific

Curriculum: CBSLSV3001- TVET Certificate 3 in Land Surveying

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Purpose statement	This core module describes the skills, knowledge and attitude required to perform angles measurements. At the end of this module, student should be able to describe surveying instruments, tools and materials used in angular measurement; perform surveying instruments set up, and perform field angles measuring.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language; Linear distance measurement; Construction drawings interpretation; Building setting out.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Describe Surveying Instruments, tools and materials in angles measurements	1.1 Surveying instruments, tools, and materials used in angles measurements are properly identified according to their use.
	1.2 Surveying instruments and tools are appropriately classified according to their functions and construction design.
	1.3 Angular measurement units are adequately identified in accordance with their unit types.
2. Perform Surveying	2.1 Surveying instruments and tools are properly selected according to the task requirements.

Instruments set up	2.2 Surveying instruments are correctly set up according to their instructional manual.
	2.3 Surveying instruments are properly adjusted according to the their types.
3. Perform field angles measurements	3.1. Types of angles are properly determined according to their applications.
	3.2. Methods of measuring angles are properly selected in accordance with their accuracies.
	3.3. Angles are accurately measured in traverse according to the conventional procedures.
	3.4. Errors of measured angles are accurately adjusted according to their types.

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Describe surveying instruments, tools, and materials in angles measurements 2. Perform Surveying instruments set up 3. Perform field angles measurements
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Learning outcome 1: Describe Surveying Instruments, Tools and Materials in angles measurements	Learning hours: 12
Indicative content	
* Surveying instruments; tools and materials used in angles measurements (Refer to ISO 8322-4, ISO 17123-2, ISO 1141) • Surveying instruments (compass, theodolite, tacheometer, levels, etc.) • Surveying tools (tripod, ranging rods, plumb bob, claw hammer, club hammer, panga, hand saw, etc.) • Materials (building lines, pegs, paper, pen, nails, theodolite batteries, etc...) * Classification of surveying instruments used in angles measurements	

• Classification of Theodolites ✓ Based on technology used (repeating theodolites, directional theodolites, electrical digital theodolites, total station) ✓ Based on movement of telescope about the horizontal axis in a vertical plane (Transit theodolites, Non transit theodolites) ✓ Based on arrangement to measure the angles (vernier theodolites, micrometer theodolites/precise optical theodolites, electronic or digital theodolites) • Classification/Types of Compass (Prismatic Compass, Surveyor's Compass, etc....) * Angular measurement units • Angular measurement units system (sexagesimal system, centesimal system, radian system) • Converting the angular measurement units system (sexagesimal to centesimal system, centesimal to sexagesimal system, sexagesimal to radian system, radian to sexagesimal system, centesimal to radian system, radian to centesimal system)	
Resources required for the learning outcome	
Equipment	Theodolite, compass, tacheometer, automatic level, dumpy level, whiteboard, projector, computer.
Materials	Notebooks, marker pen, pen, pieces of chalk, pieces of paper, photos.
Tools	Tripod, ranging rods, plumb bob, claw hammer, club hammer, panga, hand saw.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment

Learning outcome 2: Perform Surveying instruments set up	Learning hours: 13
Indicative content	

*** Selection of Surveying instruments and tools for angles measurements**

- Surveying instruments for measuring angles (compass, theodolite, tacheometer, sextant, dumpy level, automatic level)
- Surveying tools (tripod, ranging rods, plumb bob, claw hammer, club hammer, panga, hand saw)
- Theodolite set instrument up (tripod set up, unbox instrument, mount/fix instrument on head of tripod, leveling up, centering, removal of parallax)

*** Adjust surveying instruments (Refer to ISO 17025, RS ISO /IEC 17025)**

- Temporary adjustments of the instrument (setting up, centering, levelling up, focussing/removal of parallax)
- Permanent adjustments of the instrument (axis of plate levels adjustment, line of collimation adjustment, horizontal axis adjustment, adjustment of telescope level, adjustment of bubble line or altitude level, vertical axis adjustment)

Resources required for the indicative content

Equipment	Theodolite, compass, tacheometer, automatic level, dumpy level, whiteboard, projector, computer.
Materials	Notebooks, marker pen, pen, pieces of chalk, pieces of paper, photos.
Tools	Ranging rods, plumb bob, tripod.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment

Learning outcome 3: Perform field angles measurements		Learning hours: 35
Indicative content		
* Types of angles in surveying measurements • Types of angles ✓ Horizontal angles {interior and exterior angles, azimuths, whole circle bearings (W.C.B), reduced bearings (R.B), deflection angles, etc...} ✓ Vertical angles (elevation angles, depression angles, zenith angles, etc...) • Determination of angles (conversion of W.C.B. into R.B., conversion of R.B. into W.C.B., calculation of angles from bearings, calculation of bearings from angles) * Methods of measuring angles • Methods of measuring horizontal angles (ordinary method, repetition method, reiteration method) * Measure angles in traverse • Purpose of traverse • Types traverse (open traverse and closed traverse (link traverse, loop/polygonal traverse) • Steps of measuring angles * Adjusting errors of measured angles • sources of errors • types of errors • errors in angles measurements • errors in traversing (angular error, linear error, centering error) • adjust field angles		
Resources required for the indicative content		
Equipment	Theodolite, compass, tacheometer, automatic level, dumpy level, whiteboard, projector, computer.	
Materials	Notebooks, marker pen, pen, pieces of chalk, pieces of papers, photos, pegs.	
Tools	Ranging rods, plumb bob, tripod, claw hammer, club hammer, panga, ranging rods, measuring tape, hand saw, builder's square.	
Facilitation techniques	• Demonstration and simulation, Individual and group work, Practical exercise , Individualized, Trainer guided , Group discussion	

Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product based assessment
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Integrated/Summative assessment (For specific module)

Integrated situation

East gate Hotel located in Ngoma district, Kibungo sector wants to construct three (3) bungalows of different shapes and sizes on sloped terrain.

The bungalows should fulfill the following requirements:

- The rectangular bungalow of 2m×3m
- The square bungalow of 3m×3m
- The hexagon bungalow of 3m side

As a land surveyor, you are requested to lay out these bungalows by using surveying instruments; tools and materials within seven (7) hours. The architectural drawings and other needed materials, tools, and instruments/ equipment are available.

Resources

Tools	Claw hammer, club hammer, panga, ranging rods, measuring tape, hand saw builder's square.
Equipment	Theodolite
Materials/ Consumables	Building lines, pegs, pieces of paper, pen, nails, theodolite batteries.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 2: Perform	2.1. Surveying instruments tools and materials are	1. Instruments are selected.			4
		2. Tools are selected.			4
		3. Materials are selected.			4

Surveying instruments set up (40%)	properly selected.			
	2.2.Surveying instruments are correctly set up	1. Tripod is set up.		4
		2. Instrument is unboxed and boxed		4
		3. Instrument is fixed on Tripod.		4
		4. Instrument is centered.		6
		5. Instrument is leveled up.		6
		6. Parallax are removed.		4
Learning outcome 3: Perform field angles measurements (60%)	3.1. Angles are accurately measured	1. Starting point is fixed.		5
		2.Distance from instrument station to initial object is measured.		6
		3. Initial point is sighted.		4
		4. Initial angle is recorded.		5
		5. Terminal point is fixed.		5
		6. Instrument is turned.		6
		7.Distance from instrument station to terminal object is measured.		6
		8. Terminal point is sighted.		5
		9. Angle is measured.		8
		10. Bungalows are laid out.		10
Total marks		100		
Percentage Weightage		100%		
Minimum Passing line % (Aggregate): 70%				

References:

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3. Charles, D, Ghilani, P and Wolf, R. (2012). Elementary surveying. Pennsylvania State.
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9. Wolf, P, R. and Russel C. (1989). Elementary Surveying. New York: Harper and Row.
10. ISO 1141 Fabre Ropes — Polyester — 3-, 4-, 8- And 12-Strand Ropes
11. ISO 17123-2 Optics and optical instruments — Field procedures for testing geodetic and surveying instruments — Part 2: Levels
12. ISO 8322-4 Building construction — measuring instruments — Procedures for determining accuracy in use — Part 4: Theodolites
13. ISO/IEC 17025 General Requirements for the Competence of Testing And Calibration Laboratories